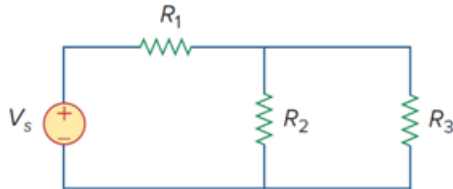


Problem

In the circuit of Fig. 2.67, a decrease in R_3 leads to a decrease of, select all that apply:

- (a) current through R_3
- (b) voltage across R_3
- (c) voltage across R_1
- (d) power dissipated in R_2
- (e) none of the above

Figure 2.67:



Step-by-step solution

Step 1 of 6

Refer to the circuit diagram in Figure 2.67 in the textbook.

Find the current flow through resistor R_3 using current division rule.

$$i_3 = i \frac{R_2}{R_2 + R_3}$$

Assume $i \frac{R_2}{R_2 + R_3} = i_o$

Consider resistor R_3 is decreased to $\frac{R_3}{2}$.

Now current i_3 becomes:

$$\begin{aligned} i_{3(\text{new})} &= i \frac{R_2}{R_2 + \frac{R_3}{2}} \\ &= 2i \frac{R_2}{2R_2 + R_3} \end{aligned}$$

Thus, current $i_{3(\text{new})}$ is increased compared to current i_o .

So, option (a) is wrong option.

Step 2 of 6

Voltage across resistor R_3 is,

$$V_3 = i_3 R_3$$

Assume the voltage across resistor R_3 as $V_3 = v_o$

Consider resistor R_3 is decreased to $\frac{R_3}{2}$.

$$\begin{aligned} V_{3(\text{new})} &= i_3 \frac{R_3}{2} \\ &= \frac{V_3}{2} \end{aligned}$$

Thus, voltage $V_{3(\text{new})}$ is decreased compared to voltage v_o . **So, option (b) is correct option.**

Step 3 of 6

Find the current flow through the resistor R_1 .

$$i_1 = \frac{V_s}{R_{eq}} \dots\dots (1)$$

Here

$$\begin{aligned} R_{eq} &= R_1 + R_2 \parallel R_3 \\ &= R_1 + \frac{R_2 R_3}{R_2 + R_3} \end{aligned}$$

Substitute $R_1 + \frac{R_2 R_3}{R_2 + R_3}$ for R_{eq} in equation (1).

$$\begin{aligned} i_1 &= \frac{V_s}{R_1 + \frac{R_2 R_3}{R_2 + R_3}} \\ &= \frac{V_s}{R_1 + \frac{R_2 R_3}{R_2 + R_3}} \end{aligned}$$

The voltage across resistor R_1 is,

$$\begin{aligned} V_1 &= R_1 i_1 \\ &= R_1 \left(\frac{V_s}{R_1 + \frac{R_2 R_3}{R_2 + R_3}} \right) \end{aligned}$$

Step 4 of 6

Consider resistor R_3 is decreased to $\frac{R_3}{2}$.

$$\begin{aligned}
 V_1(\text{new}) &= R_1 i_1 \\
 &= R_1 \left(\frac{V_s}{R_1 + \frac{R_2 \frac{R_3}{2}}{R_2 + \frac{R_3}{2}}} \right) \\
 &= \frac{V_s R_1}{R_1 + \frac{R_2 R_3}{2R_2 + R_3}}
 \end{aligned}$$

From the expression it is clear that denominator reduced which leads to overall voltage $V_1(\text{new})$ increases.

Thus, the voltage across resistor R_1 increases by decreasing resistor R_3 .

So, option (c) is wrong option.

Step 5 of 6

Power dissipated in R_2 .

$$\begin{aligned}
 P_{R_2} &= i_2 v_2 \\
 &= i_2 v_3 \quad \text{since } v_3 = v_2 \\
 &= i_2 (R_3 i_3) \quad \text{since } v_3 = R_3 i_3 \\
 &= \left(\frac{i_1 R_3}{R_2 + R_3} \right) i_3 R_3 \quad \text{since } i_2 = \frac{i_1 R_3}{R_2 + R_3} \\
 &= \left(\frac{V_s}{R_1 + \frac{R_2 R_3}{R_2 + R_3}} \right) \left(\frac{R_3^2 i_3}{R_2 + R_3} \right) \quad \text{since } i_1 = \frac{V_s}{R_1 + \frac{R_2 R_3}{R_2 + R_3}}
 \end{aligned}$$

Consider resistor R_3 is decreased to $\frac{R_3}{2}$.

$$\begin{aligned}
 P_{R_2(\text{new})} &= \frac{V_s}{R_1 + \frac{R_2 \frac{R_3}{2}}{R_2 + \frac{R_3}{2}}} \left(\frac{\frac{R_3}{2}}{R_2 + \frac{R_3}{2}} \right) i_3 \frac{R_3}{2} \\
 &= \frac{V_s}{R_1 + \frac{R_2 R_3}{2R_2 + R_3}} \left(\frac{\frac{R_3}{2}}{R_2 + \frac{R_3}{2}} \right) i_3 \frac{R_3}{2} \\
 &= \left(\frac{V_s}{R_1 + \frac{R_2 R_3}{2R_2 + R_3}} \right) \left(\frac{i_3 R_3^2}{4(2R_2 + R_3)} \right)
 \end{aligned}$$

It is clear that $P_{R_2(\text{new})}$ is decreased compared to P_{R_2} . **So, option (d) is correct option.**

Step 6 of 6

Therefore, voltage across R_3 and power dissipated in R_2 decreases with resistor R_3 decrease. **So correct options are (b) and (d).**

